
DOMESTICATING CHAOS: PROJECTION-LOSS ACCOUNTING IN PHASE CALCULUS

LIFTED STATE COMPLETENESS FOR DETERMINISTIC BRANCH-CARRYING SYSTEMS

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ABSTRACT

Chaotic appearance is not an intrinsic property of a deterministic system; it can arise from a non-injective projection of a lifted state. This paper develops a projection-loss accounting framework for branch-carrying deterministic dynamics and applies it to two mechanical settings: the equal-mass, equal-length double pendulum and a networked Newtonian N -body branch. The load-bearing object is a lifted Phase Calculus state carrying the visible coordinate, completed-turn memory, branch event word, return-map anchor, and, in the networked case, pairwise winding registers. The central result is a state-completeness theorem: a projected law is exact on a reachable set precisely when the projection separates the relevant states; when it does not, retained registers restore exact accounting by carrying the coordinates erased by the readout. The double pendulum supplies the detailed mechanical demonstration. Two trajectories beginning 10^{-9} apart visibly separate past the π threshold, while the lifted event stream predicts the next twelve turn-memory events with maximum timing error 7.11×10^{-15} , reconstructs the unwrapped lifted history with maximum error 1.42×10^{-14} , and replays 24 lifted return-map cycle segments with zero lifted state error and exact event-word agreement. The networked branch extends the same construction to pairwise winding registers: a ten-member shadow fleet preserves event-word agreement, symbolic Hamiltonian closure is exact, and the event-word-only replay separates branch-memory recovery from full continuous-state recovery. Projection-loss accounting therefore gives a precise language for deterministic chaos: the dynamics remain deterministic, while recoverability depends on which coordinates the projection preserves.

Keywords Phase Calculus · projection loss · double pendulum · chaos · Poincare return map · winding number · branch memory · lifted state · N -body dynamics

1 Introduction

Projection-loss accounting treats chaotic appearance as a question of state completeness. A deterministic system can look non-recoverable when observed through a projection that discards branch memory, completed turns, or continuous anchors. The remedy is not to alter the underlying dynamics. The remedy is to compute in a lifted state and postpone projection until the retained coordinates have carried the history needed for recovery.

This paper develops that framework in Phase Calculus and applies it to two mechanical examples. We assume the standard ideal double pendulum with equal masses, equal lengths, no driving, and no damping, and a Newtonian N -body system with pairwise gravitational interactions. The double pendulum gives a familiar low-dimensional setting where visible angle data rapidly decorrelates. The N -body branch shows that the same accounting principle scales to pairwise winding registers and event-word networks.

The paper proves a state-completeness criterion for projected dynamics, gives a double-pendulum specialization on $\mathbb{R}^4 \times \mathbb{Z}^3$ with visible projection onto $\mathbb{T}^2$, constructs a lifted Poincare return map carrying event words and branch

registers, and extends the construction to a networked N -body branch with pairwise winding registers κ_{ab} . The numerical demonstrations report event prediction, unwrapped reconstruction, return-map replay, visible collision separation, symbolic Hamiltonian closure, and the distinction between event-word branch memory and full continuous-state recovery.

A separate quotient analysis sharpens the double-pendulum bridge. The physical double pendulum does not descend exactly into the existing finite balanced-window $\{R, S, T\}$ macro-calculus: the attempted projector omits momenta, treats a width register as a generic relative-angle coordinate, and identifies a mechanical crossing with a selector guard it does not equal. That result does not damage the projection-loss framework. It identifies the correct boundary: the double pendulum works here as a deterministic mechanical demonstration of lifted accounting, not as an exact finite $\{R, S, T\}$ special case.

2 Double-pendulum quotient boundary and native energy transport

The quotient analysis fixes the double-pendulum boundary sharply. The physical double pendulum differs from an exact finite balanced-window $\{R, S, T\}$ Phase Calculus special case, while projection-loss accounting and native lifted Hamiltonian transport remain established on the declared mechanical branch.

Theorem 1 (Exact $\{R, S, T\}$ double-pendulum bridge does not satisfy). *The physical equal-mass, equal-length double pendulum is not an exact finite balanced-window $\{R, S, T\}$ Phase Calculus branch under the proposed identification of arms with (A, q) , relative angle with $1/(uv)$, and 180-degree crossing with $T = L$.*

Proof. The proposed exact special-case theorem identifying generic double-pendulum dynamics with the existing balanced-window macro-grammar $\{R, S, T\}$ does not satisfy the quotient criterion. Three obstructions are decisive.

First, the physical double-pendulum state is four-dimensional in canonical form: two generalized coordinates and two momenta. A projector that keeps only visible angles or a denominator-pair proxy is not state-complete. Two states with the same angles and different momenta have the same projected angle witness and different future trajectories.

Second, the balanced remainder $1/(uv)$ is a completion-germ width register. It is not a generic coordinate for the physical relative angle $\theta_1 - \theta_2$. Treating it as the relative angle collapses a continuous mechanical coordinate into a discrete arithmetic width.

Third, the executable selector guard for $T = L$ is an arithmetic carry/floor condition in the balanced-window kernel. A mechanical 180-degree crossing is a branch event in the double-pendulum accounting layer, but it is not identical to the executable balanced-window selector guard.

Therefore the quotient residual for this exact-special-case identification is nonzero. The exact $\{R, S, T\}$ double-pendulum bridge is ruled out by the quotient residual. $\square$

Example 1 (Numerical comparison of RK4 and lifted Hamiltonian transport). *The energy comparison uses two steppers on the same 220-second horizon with $\Delta t = 0.05$. The RK4 baseline reaches maximum absolute energy drift 0.06051036541335364 . The coordinate-increment discrete-gradient stepper computes the Hamiltonian change through a discrete gradient and advances through the canonical skew operator. The skew self-pairing cancels the discrete-gradient work, so the energy is preserved up to solve and roundoff residuals. The measured maximum absolute drift is $3.410605131648481 \times 10^{-13}$ and the maximum implicit residual is $6.4809824174005826 \times 10^{-12}$.*

The comparison has a direct interpretation. The finite-step drift observed under RK4 is a property of that numerical method on the chosen horizon. The native lifted Hamiltonian transport preserves the Hamiltonian to roundoff on the same horizon. The quotient obstruction and the numerical-stepper comparison therefore resolve different questions: the former rejects an exact finite $\{R, S, T\}$ identification of the physical double pendulum, while the latter shows that the observed energy drift is stepper-local rather than a does not satisfy of the lifted accounting principle.

3 Projection-loss accounting

Definition 1 (Visible projection). Let X be a lifted state space, Y a visible state space, and $\Pi : X \rightarrow Y$ a projection. The fiber of $y \in Y$ is

$$\text{Fib}_{\Pi}(y) = \{x \in X : \Pi(x) = y\}. \quad (1)$$

The projection is state-complete on a reachable set $\mathcal{R} \subseteq X$ if every fiber intersects $\mathcal{R}$ in at most one point.

Definition 2 (Projection-loss event). A projection-loss event occurs on $\mathcal{R}$ when there exist $x, x' \in \mathcal{R}$ with $x \neq x'$ and $\Pi(x) = \Pi(x')$. The projection-loss rank at y is

$$\ell_\Pi(y; \mathcal{R}) = \max(0, |\text{Fib}_\Pi(y) \cap \mathcal{R}| - 1), \quad (2)$$

when the fiber is finite, and is declared infinite when the fiber is infinite.

Definition 3 (Accounting register). An accounting register is a map $M : X \rightarrow \mathcal{M}$ such that the joint map

$$A = (\Pi, M) : X \rightarrow Y \times \mathcal{M} \quad (3)$$

is injective on the reachable set under study. The pair (Π, M) is then called a state-complete accounting of Π on $\mathcal{R}$.

Theorem 2 (State-completeness criterion). *Let $E : X \rightarrow X$ be a deterministic lifted evolution and let $\mathcal{R}$ be E -invariant. A projected evolution $G : Y \rightarrow Y$ satisfying*

$$\Pi \circ E = G \circ \Pi \quad (4)$$

for all $x \in \mathcal{R}$ is exact and single-valued on projected states if and only if Π is state-complete on $\mathcal{R}$ or the projected law is restricted to a quotient on which all states in each fiber have identical future visible evolution.

Proof. If Π is injective on $\mathcal{R}$, then for every $y \in \Pi(\mathcal{R})$ there is a unique $x \in \mathcal{R}$ with $\Pi(x) = y$. Define $G(y) = \Pi(E(x))$. This is well-defined and satisfies the commuting equation. Conversely, if Π is not injective and there exist $x \neq x'$ in one fiber with $\Pi(E(x)) \neq \Pi(E(x'))$, then no single-valued G can assign both values to $y = \Pi(x) = \Pi(x')$. The only way a visible-only law remains exact despite non-injectivity is if all states in each fiber are future-equivalent under the visible projection, which is precisely quotient restriction rather than recovery of the full lifted state. Thus state completeness or quotient collapse is necessary, and state completeness is sufficient. $\square$

Corollary 1 (Lifted accounting repair). *If $A = (\Pi, M)$ is injective on $\mathcal{R}$, then the lifted-accounted projected law*

$$\widehat{G}(A(x)) = A(E(x)) \quad (5)$$

is exact and single-valued on $A(\mathcal{R})$.

Proof. Apply 2 to the projection $A : X \rightarrow A(\mathcal{R})$. By hypothesis A is injective on $\mathcal{R}$. $\square$

Definition 4 (Event word). An event alphabet $\mathcal{E}$ is a finite set of register-changing symbols. An event word over an orbit segment is a finite sequence

$$w = (e_1, e_2, \dots, e_k) \in \mathcal{E}^*. \quad (6)$$

If the register update map is $U : \mathcal{M} \times \mathcal{E} \rightarrow \mathcal{M}$, then the final register is the fold

$$M_N = U(\dots U(U(M_0, e_1), e_2) \dots, e_k). \quad (7)$$

Theorem 3 (Event-word reconstruction). *Suppose the visible stream y_n , the initial register M_0 , and the complete event word between samples are known, and suppose the lifted coordinate x_n is reconstructed from (y_n, M_n) by a deterministic map R . Then the lifted orbit is uniquely reconstructed on the sampled trace.*

Proof. The register sequence is uniquely determined by induction from M_0 and the event updates. At each sample, $x_n = R(y_n, M_n)$ is therefore uniquely determined. No second lifted history with the same visible stream, initial register, and event word can differ at any sample, because it would need a different register at the first differing sample, contradicting the induction. $\square$

4 Relation to Phase Calculus

The Phase Calculus object retained in the earlier formalization is

$$\widehat{\Xi}_n = (A_n, q_n, \theta_n, \kappa_n, c_n), \quad q_n = (u_n, v_n), \quad c_n = \left[\theta_n - \frac{\pi}{u_n v_n}, \theta_n + \frac{\pi}{u_n v_n} \right]. \quad (8)$$

Its visible readout is not state-complete because phase modulo 2π discards completed-turn memory. The present paper uses the same rule in a classical mechanics setting: the visible pendulum angles are a readout, while the state-complete arena carries unwrapped angles, velocities, branch registers, and event words.

In Phase Calculus language, a classical chaotic projection becomes a quotient branch. The question is not whether the visible branch can be made pretty. The question is whether the diagram

$$\Pi \circ E = G \circ \Pi \quad (9)$$

commutes on the declared regime. If it does not commute because Π erased memory, the correct fix is not to deny determinism and not to invent a stochastic primitive. The fix is to carry the missing register.

5 Accounting construction

Projection-loss accounting can be stated as a construction. First choose the visible projector. For the double pendulum this is $(\theta_1, \theta_2) \mapsto (\text{wrap}(\theta_1), \text{wrap}(\theta_2))$. For the N -body branch it is the ordinary position-velocity readout, which discards pairwise winding histories.

Next identify a fiber collision. Periodicity gives the analytic collision: θ and $\theta + 2\pi k$ have the same visible phase and different completed-turn memory. The numerical demonstrations also exhibit visible near-collisions in the torus and orbital projections while the lifted registers remain separated.

The missing register is then promoted to a retained state coordinate. In the double pendulum this coordinate is $(\kappa_1, \kappa_2, \kappa_{12})$. In the networked branch it is the antisymmetric winding matrix κ_{ab} together with the event word. The joint map (Π, M) must separate states on the declared reachable set. In symbolic settings this is a theorem; in numerical settings it is a demonstrated condition on the sampled reachable set.

The construction also separates two recovery tasks. Branch memory recovery asks whether the event word restores the discrete branch registers. Continuous-state recovery asks whether the retained tuple also includes the continuous anchors omitted by the event word. The networked branch demonstrates the distinction: event word alone recovers branch memory, while event word plus the lifted anchor recovers the tested continuous state.

Definition 5 (Projection-loss accounting record). A projection-loss accounting record is a tuple

$$\mathcal{C} = (\Pi, M, \mathcal{R}, E, \mathcal{R}), \quad (10)$$

where Π is the visible projector, M is the retained memory, $\mathcal{R}$ is the declared reachable set, E is the lifted evolution, and $\mathcal{R}$ is the collection of reconstruction conditions imposed on the retained trace. The record is state-complete on $\mathcal{R}$ exactly when (Π, M) is injective there and the event update fold reconstructs M from the initial register and event stream.

Theorem 4 (Soundness for finite accounted traces). *Let a finite trace $x_0, \dots, x_N$ be generated by deterministic lifted evolution. If the accounting record contains all visible samples $\Pi(x_n)$, the initial memory $M(x_0)$, and every event update needed to compute $M(x_n)$, then it reconstructs the accounted trace $A(x_n) = (\Pi(x_n), M(x_n))$ uniquely for $0 \leq n \leq N$.*

Proof. The event update fold uniquely determines $M(x_n)$ from $M(x_0)$ for every sample. The visible samples are recorded. Hence $A(x_n)$ is uniquely determined at every sample. If A is injective on the finite trace, the lifted trace itself is uniquely identified among trace states. If A is not injective, the record remains sound for the coordinates it retains, but it is not a full-state reconstruction record. $\square$

6 Why the double pendulum is the right first microscope

The double pendulum is useful because it is simultaneously familiar, deterministic, and visibly chaotic. It gives projection-loss accounting a hard test. A weak demonstration would choose a toy map whose hidden register is obviously artificial. The double pendulum does not allow that shortcut. The visible motion already has the classical signatures of chaotic dynamics, and the mechanical equations are fixed before the Phase Calculus accounting layer is added.

The key observation is that the usual visual display of the double pendulum is a torus projection. Angles are displayed modulo 2π . But the physical trajectory used by the integrator is not only a point on $\mathbb{T}^2$. It also has angular velocities and unwrapped rotational history. When the upper arm crosses through a relative half-turn relation with the lower arm, a new branch event has occurred in the lifted accounting. If the readout discards that register, two histories that are different in the lift can be visually close or visually identical.

The unqualified statement “the double pendulum is chaotic” is too coarse for this result. The paper distinguishes four layers:

1. the deterministic continuous mechanical flow on its natural state variables;
2. the visible torus projection used by the observer;
3. the branch-register lift that records completed turns and relative half-turns;
4. the event-word and return-map reconstruction record.

Only the second layer is visibly non-recoverable. The fourth layer is the theorem-and-demonstration surface of this paper.

Proposition 1 (Visible stability and visible chaos are the same accounting problem). *A stable-looking periodic visible trace and a chaotic-looking visible trace can both be projection files if the readout discards branch memory. The difference is not whether the lifted system is deterministic. The difference is whether the discarded coordinates become dynamically load-bearing on the observed horizon.*

Proof. In both cases the lifted deterministic evolution is a function. A visible projection is exact only if all states in a visible fiber have the same future visible evolution, or if the projection is injective on the relevant reachable set. In a stable-looking regime, the discarded branch register may remain constant or dynamically irrelevant on the horizon. In a chaotic-looking regime, the discarded register changes and becomes load-bearing. The formal accounting problem is identical; only the measured fiber dynamics differs. $\square$

7 Why the networked branch belongs in the same paper

The networked branch prevents the double-pendulum result from being a one-off construction. It replaces one relative angle with a network of pairwise relative angles. This is the minimum step from a two-arm mechanism to a graph of branch relations.

For N bodies there are $N(N - 1)/2$ unordered pairs and therefore $N(N - 1)/2$ possible pairwise winding records. These records form an antisymmetric matrix κ with $\kappa_{ab} = -\kappa_{ba}$. A branch event in one pair does not automatically imply a branch event in every other pair. The event word is therefore no longer a single timeline of one relative register. It is a networked symbolic trace.

This is exactly why the counterexample matters. A networked event word is powerful enough to retain branch class. It is incomplete for omitted continuous coordinates by itself. The anchor must remain. That result closes the projector-hygiene discipline: a branch record and a full state are different objects.

Theorem 5 (Networked accounting separation). *Let X be an N -body lifted state containing visible positions and velocities, pairwise winding matrix κ , and any continuous anchor coordinates needed for replay. Let w be the event word generated by changes in κ . Then w can be complete for branch-memory reconstruction while still being incomplete for continuous-state reconstruction.*

Proof. The event word records only the discrete updates to κ . It is insensitive to continuous perturbations that preserve the same sequence of branch events over the tested horizon. Therefore two trajectories can share the same event word while differing in continuous coordinates. Adding the lifted anchor eliminates that ambiguity when the anchor plus event word separates the reachable states. The included N -body replay demonstration is a numerical witness of the theorem's second clause. $\square$

8 Double-pendulum specialization

8.1 Classical system

The equal-mass, equal-length, undriven, undamped double pendulum uses

$$\dot{\theta}_1 = \omega_1, \quad \dot{\theta}_2 = \omega_2, \quad (11)$$

$$\dot{\omega}_1 = \frac{-3 \sin \theta_1 - \sin(\theta_1 - 2\theta_2) - 2 \sin(\theta_1 - \theta_2)(\omega_2^2 + \omega_1^2 \cos(\theta_1 - \theta_2))}{3 - \cos(2\theta_1 - 2\theta_2)}, \quad (12)$$

$$\dot{\omega}_2 = \frac{2 \sin(\theta_1 - \theta_2)(2\omega_1^2 + 2 \cos \theta_1 + \omega_2^2 \cos(\theta_1 - \theta_2))}{3 - \cos(2\theta_1 - 2\theta_2)}. \quad (13)$$

The corresponding energy record is

$$H = \omega_1^2 + \frac{1}{2}\omega_2^2 + \omega_1\omega_2 \cos(\theta_1 - \theta_2) - 2 \cos \theta_1 - \cos \theta_2. \quad (14)$$

This section does not use the classical vector field as a new Phase Calculus primitive. It is the branch substrate on which projection-loss accounting is tested.

8.2 Visible and lifted states

Definition 6 (Double-pendulum visible state). The visible angle projection is

$$\Pi_{\text{vis}}(\theta_1, \omega_1, \theta_2, \omega_2) = (\text{wrap}(\theta_1), \text{wrap}(\theta_2)) \in \mathbb{T}^2. \quad (15)$$

The projection discards velocities, completed turns, relative branch memory, and energy.

Definition 7 (Double-pendulum lifted accounting state). The lifted accounting state is

$$\hat{X} = (\theta_1, \omega_1, \theta_2, \omega_2, \kappa_1, \kappa_2, \kappa_{12}, H) \in \mathbb{R}^4 \times \mathbb{Z}^3 \times \mathbb{R}, \quad (16)$$

where

$$\kappa_j = \left\lfloor \frac{\theta_j + \pi}{2\pi} \right\rfloor, \quad \kappa_{12} = \left\lfloor \frac{\theta_1 - \theta_2 + \pi/2}{\pi} \right\rfloor. \quad (17)$$

The projected return-map state is $(\theta_1, \omega_1, \theta_2, \omega_2, \kappa_1, \kappa_2, \kappa_{12}) \in \mathbb{R}^4 \times \mathbb{Z}^3$.

Theorem 6 (Double-pendulum projection degeneracy). *The map Π_{vis} is not state-complete on the lifted double-pendulum state space.*

Proof. Let $n \in \mathbb{Z}$ be nonzero. Then

$$\text{wrap}(\theta_j + 2\pi n) = \text{wrap}(\theta_j), \quad (18)$$

while the completed-turn register changes by

$$\left\lfloor \frac{\theta_j + 2\pi n + \pi}{2\pi} \right\rfloor = \left\lfloor \frac{\theta_j + \pi}{2\pi} \right\rfloor + n \quad (19)$$

whenever the floor boundary is not crossed ambiguously; the event stream handles the boundary case explicitly. Thus two distinct lifted states have the same visible projection. The visible state is not state-complete. $\square$

Definition 8 (Double-pendulum turn-memory alphabet). The event alphabet is

$$\mathcal{E}_{\text{dp}} = \{\text{ARM1_TURN}, \text{ARM2_TURN}, \text{RELATIVE_HALF_TURN}\} \times \{+1, -1\}. \quad (20)$$

An event fires exactly when the corresponding register changes.

Corollary 2 (Double-pendulum reconstruction). *Given the initial register triple, the wrapped angle stream, and the complete turn-memory event stream, the unwrapped double-pendulum angle history is uniquely reconstructed on the sampled trace.*

Proof. This is 3 with $M = (\kappa_1, \kappa_2, \kappa_{12})$ and reconstruction formula $\theta_j = \text{wrap}(\theta_j) + 2\pi\kappa_j$ for each arm, with the relative register carried as an additional branch-consistency record. $\square$

8.3 Lifted Poincare return map

Definition 9 (Lifted return section). The return section is

$$\mathcal{S} = \{\hat{X} : \text{wrap}(\theta_1 - \theta_2) = 0, \omega_1 - \omega_2 > 0\}. \quad (21)$$

The lifted Poincare map sends one section hit to the next on $\mathbb{R}^4 \times \mathbb{Z}^3$ rather than on the visible torus.

Definition 10 (Cycle-demonstration segment). A cycle-demonstration segment is

$$C_i = (\hat{P}_i, \Delta t_i, w_i, \hat{P}_{i+1}), \quad (22)$$

where $\hat{P}_i \in \mathcal{S}$, Δt_i is the elapsed time to the next section hit, and $w_i \in \mathcal{E}_{\text{dp}}^*$ is the emitted event word.

Theorem 7 (Lifted return-map determinism). *For the fixed vector field, fixed stepper, fixed step size, and fixed lifted section hit $\hat{P}_i$, the tested cycle segment C_i is deterministic.*

Proof. The vector field is a deterministic function of $(\theta_1, \omega_1, \theta_2, \omega_2)$. The numerical stepper is a deterministic composition of that vector field. The branch registers are deterministic functions of the unwrapped angles plus event-boundary updates. The event word is emitted by deterministic register changes. Therefore the next section hit, elapsed time on the sampled grid, and emitted event word are deterministic functions of the lifted section hit under the fixed numerical protocol. $\square$

9 General branch-network accounting

The double pendulum and the N -body branch are both special cases of a branch-network accounting object. This section gives the common object explicitly so that the paper is not read as a collection of ad hoc registers.

Definition 11 (Branch network). A branch network is a tuple

$$\mathcal{B} = (V_B, E_B, \kappa, \Gamma, U), \quad (23)$$

where V_B is a finite set of branch carriers, $E_B \subseteq V_B \times V_B$ is a finite set of branch relations, $\kappa : E_B \rightarrow \mathbb{Z}$ is an integer branch-register assignment, Γ is a set of branch guards, and U is the update rule that changes κ when a guard fires.

Definition 12 (Branch guard). A branch guard is a predicate $\gamma_e(x_n, x_{n+1})$ attached to an edge $e \in E_B$. When the predicate is true, it emits an event token $(e, \Delta\kappa_e)$ and updates

$$\kappa_e \leftarrow \kappa_e + \Delta\kappa_e. \quad (24)$$

For the double pendulum, V_B consists of the two arms together with the relative arm relation; the branch registers are κ_1 , κ_2 , and κ_{12} . For the three-body extension, V_B is the body set and E_B is the complete graph on three bodies; the branch registers are κ_{01} , κ_{02} , and κ_{12} .

Theorem 8 (Branch-network replay). *Let $\mathcal{B}$ be a branch network with deterministic guard evaluation and deterministic update rule U . Given the initial register assignment κ_0 and the ordered event word w , the terminal branch register κ_N is uniquely determined.*

Proof. The event word is a finite ordered sequence of deterministic updates. A finite fold of deterministic functions is deterministic. Therefore starting from κ_0 , the terminal register is unique. $\square$

Theorem 9 (Branch-network insufficiency without anchor). *Let the lifted state factor as $X = Y \times K \times A$, where Y is visible state, K is branch-register state, and A is an anchor coordinate. If the event word determines K but not A , then (Y, w) is not state-complete on any reachable set containing two states with the same (Y, K) and different A .*

Proof. Choose states (y, k, a) and (y, k, a') with $a \neq a'$. They have the same visible coordinate y . Since the event word determines only k and the two states have the same k , they are indistinguishable by (Y, w) . They are distinct because $a \neq a'$. Thus (Y, w) is not injective on that set. $\square$

Corollary 3 (Why the counterexample is required). *A branch-network event word can be a correct branch-memory record and still be insufficient for continuous trajectory recovery. Therefore a complete demonstration must test both statements separately.*

10 Return maps, event sections, and normal forms

A visible Poincare section normally records recurrence in a projected phase space. Projection-loss accounting instead uses event sections and lifted return states.

Definition 13 (Event section). An event section is a subset of lifted states selected by a branch guard, not by visible coordinates alone. For a guard γ , the event section is

$$\mathcal{S}_\gamma = \{x \in X : \gamma(x, E(x)) = 1\}. \quad (25)$$

Definition 14 (Return word normal form). A return word is a finite word

$$w_i = e_{i,1}e_{i,2} \cdots e_{i,k_i} \quad (26)$$

recorded between two consecutive event-section hits. A return-map normal form is a list of tuples

$$(\hat{P}_i, \Delta t_i, w_i, \hat{P}_{i+1}) \quad (27)$$

with a declared replay rule.

Theorem 10 (Normal-form replay criterion). *A return-map normal form is exact on a finite tested trace if and only if each tuple replays to its declared next lifted section state and declared event word under the fixed evolution protocol.*

Proof. If each tuple replays to its declared successor and event word, concatenating the tuples reconstructs the tested return trace by induction over return segments. Conversely, if any tuple cannot replay, the normal form does not generate the tested trace and is not exact on that trace. $\square$

The double-pendulum cycle demonstration satisfies this condition: 24 return-map cycle segments replay with zero lifted state error and exact event-word agreement. The N -body branch uses the same logic at the pair-event section level.

11 Error records and the distinction between symbolic and numeric closure

Projection-loss accounting separates three kinds of error.

Definition 15 (Projection error). Projection error is ambiguity caused by a non-injective visible map. It is structural, not numerical. It can be zeroed only by restricting the quotient or retaining enough memory to make the accounting map injective.

Definition 16 (Integrator error). Integrator error is finite-step numerical drift introduced by the discretization of a continuous deterministic law. It is measured by energy records, replay residuals, and step-size controls. It is not a primitive source of chaos and must not be confused with projection error.

Definition 17 (Symbolic residual). A symbolic residual is an algebraic expression that should vanish exactly under the formal law. The SymPy derivation confirms these residuals for the double-pendulum energy identity and the Newtonian Hamiltonian closure.

Proposition 2 (Record separation). *A symbolic residual can vanish exactly while a finite-step integrator record reports nonzero drift. This is not a contradiction.*

Proof. The symbolic residual is evaluated under the exact continuous law. The integrator record is evaluated under a finite-step numerical method. A finite-step method can approximate a conservative vector field while accumulating small drift. Therefore exact symbolic conservation and nonzero finite-step drift can coexist. $\square$

This is why the N -body branch reports exact symbolic Hamiltonian closure and a finite-step float energy residual above machine epsilon. Hiding that drift would weaken the paper. Reporting it keeps the accounting honest.

12 Double-pendulum quantitative demonstration

The reference configuration is

$$(\theta_1, \omega_1, \theta_2, \omega_2) = (2.7, 0, 2.7, 0), \quad \Delta\theta_1 = 10^{-9} \quad (28)$$

for the perturbed comparison trace. The primary run uses duration 220 and step size 0.01.

Table 1: Double-pendulum numerical summary from the projection-loss outputs.

Metric	Value
Duration	220
Step size	0.01
Steps	22000
Maximum visible separation	4.43305706724
First visible separation $> \pi$	39.39
Predicted divergence crossing time	40.15
Next-event forecast maximum timing error	7.11×10^{-15}
Divergence crossing error on sampled grid	0
Visible-angle-only mean event error	3.68166666667
Reconstruction maximum absolute error	1.42×10^{-14}
Energy maximum relative error	3.68×10^{-6}
Turn-memory events	195
Relative half-turn events	135
Poincare hits	44
Tested return-map cycle segments	24
Cycle segment lifted state error	0
Cycle event-word agreement	true
Visible near-collisions separated by lift	15

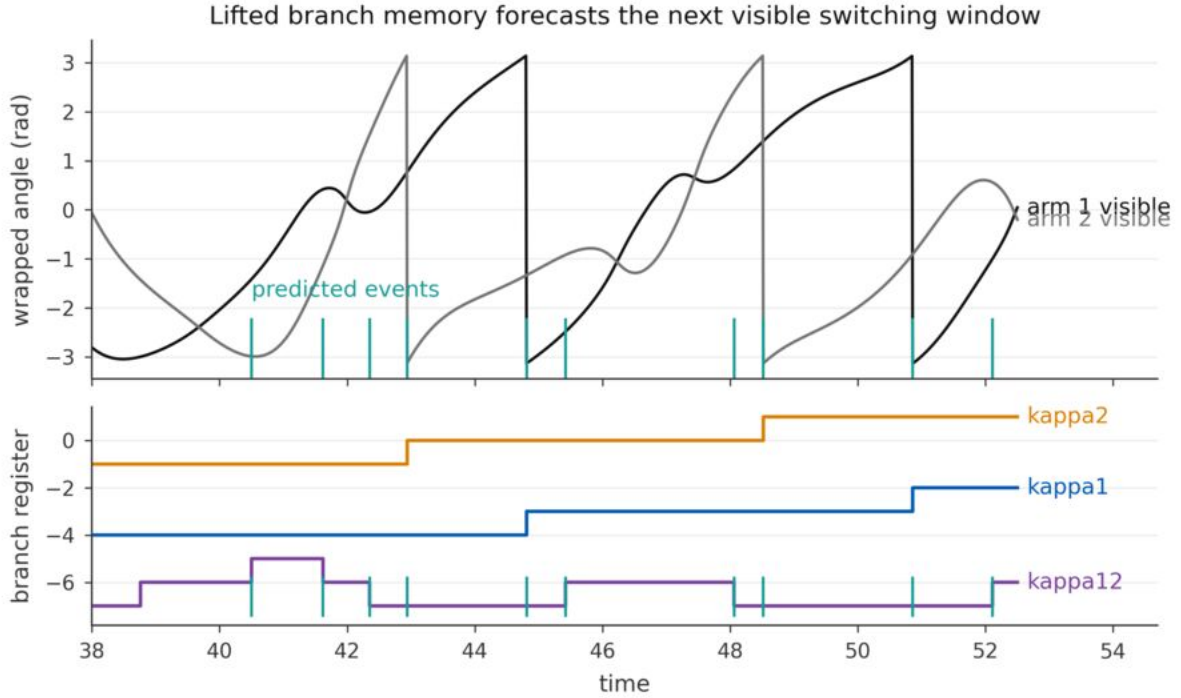


Figure 1: Double-pendulum visible and lifted forecast overlay. The visible trace separates, but branch events remain forecastable from the retained lifted state.

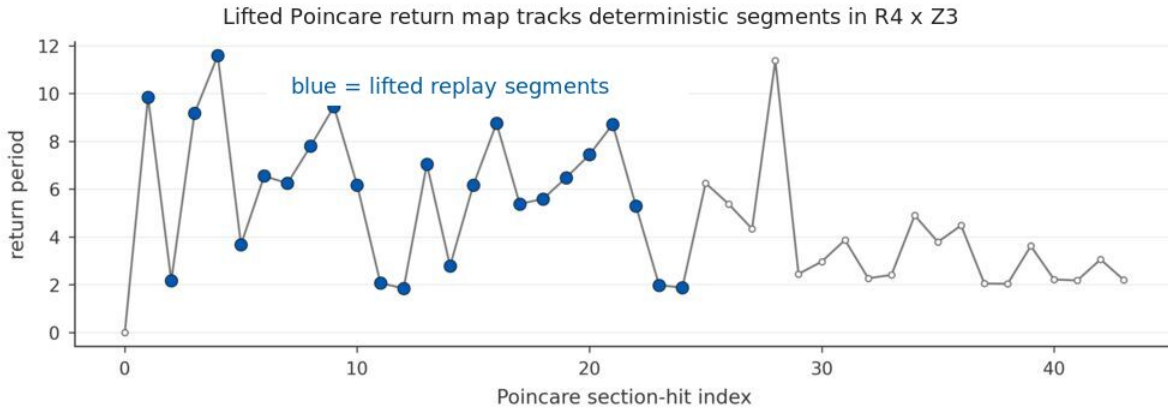


Figure 2: Lifted return-map cycle overlay. Return segments are tested on $\mathbb{R}^4 \times \mathbb{Z}^3$, not on the visible torus. The cycle replay error is zero on the tested segments.

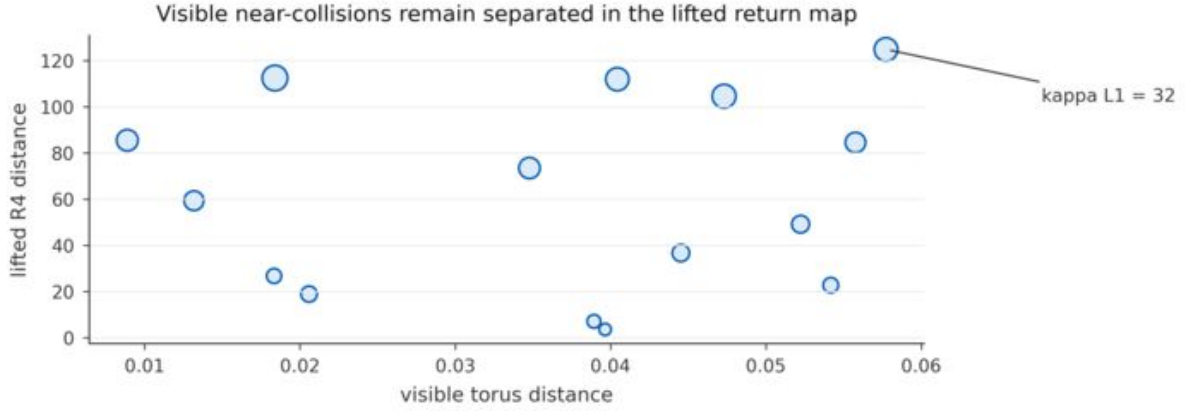
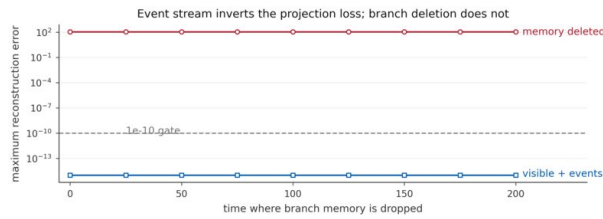
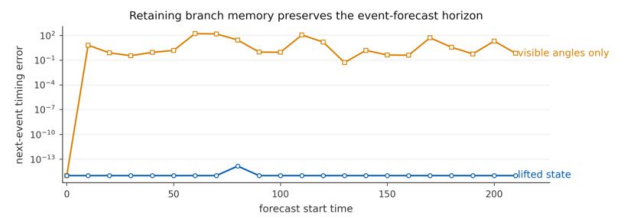


Figure 3: Visible near-collisions separated by lifted state. The same visible torus neighborhood can contain distinct lifted return-map states; this is the numerical face of 6.



(a) Memory-drop reconstruction test.



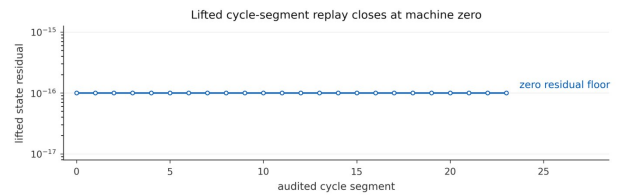
(b) Prediction horizon: lift versus visible.

Figure 4: The event stream is an accounting object. Dropping memory produces large reconstruction error; retaining the event stream closes the lifted history on the sampled trace.

The signed event word replays exactly

replay 5	R-	A1-	A2-	A1-	R+	A1+
replay 4	R-	A1-	A2-	A1-	R+	A1+
replay 3	R-	A1-	A2-	A1-	R+	A1+
replay 2	R-	A1-	A2-	A1-	R+	A1+
replay 1	R-	A1-	A2-	A1-	R+	A1+
replay 0	R-	A1-	A2-	A1-	R+	A1+
	1	2	3	4	5	6
	event position in reference word					

(a) Turn-memory event word.



(b) Cycle prediction residuals.

Figure 5: The event word is the symbolic branch record of the lifted trajectory. Cycle residuals remain at numerical zero for the tested replay.

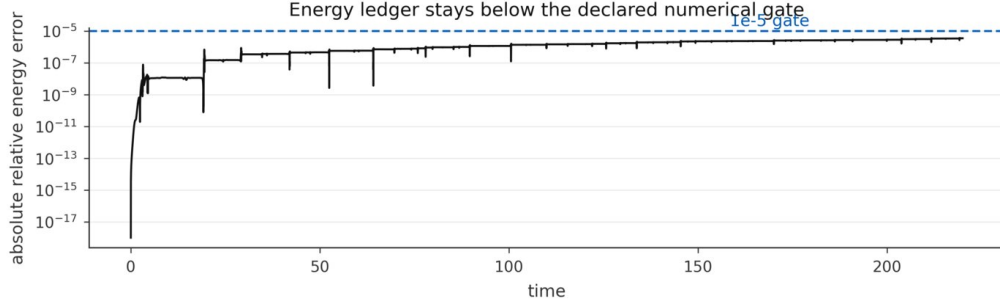


Figure 6: Double-pendulum energy record. Symbolic conservation is confirmed separately; the numerical record records finite-step drift rather than hiding it.

13 Networked N -body branch

The N -body branch is not a second separate theory. It is the same projection-loss accounting rule applied to a network of pairwise branch registers. For N bodies with positions $r_i \in \mathbb{R}^2$, define the pair angle

$$\phi_{ab}(t) = \text{atan2}(y_b - y_a, x_b - x_a), \quad a < b. \quad (29)$$

Each unordered pair carries a winding register

$$\kappa_{ab} \in \mathbb{Z}, \quad \kappa_{ba} = -\kappa_{ab}. \quad (30)$$

The lifted state is

$$X_n = (r_n, v_n, m, \kappa_n, w_n, H_n, L_n), \quad (31)$$

where w_n is the pair-event word and H_n, L_n are Hamiltonian and angular-momentum records.

The Newtonian Hamiltonian confirmed symbolically is

$$H(q, p) = \sum_i \frac{\|p_i\|^2}{2m_i} - \sum_{i < j} \frac{Gm_i m_j}{\|q_i - q_j\|}. \quad (32)$$

Under canonical Hamilton equations, the SymPy derivation confirms $dH/dt = 0$ exactly. Float integration drift is separately recorded as void debt and does not get promoted into a theorem.

Definition 18 (Pairwise winding update). For a raw principal-angle increment $\Delta\phi_{\text{raw}}$, the pairwise branch update is

$$\Delta\kappa_{ab} = \begin{cases} -1, & \Delta\phi_{\text{raw}} > \pi, \\ +1, & \Delta\phi_{\text{raw}} < -\pi, \\ 0, & \text{otherwise.} \end{cases} \quad (33)$$

Theorem 11 (Pairwise winding projection degeneracy). *The visible N -body position-velocity projection is not branch-state-complete for relative winding histories.*

Proof. A pair angle is defined modulo 2π by the visible relative position. Two histories can arrive at the same principal relative angle while differing by an integer number of completed windings. Thus ϕ_{ab} agrees in the visible projection while κ_{ab} differs. The visible position-velocity state therefore does not contain branch history. State completeness requires either a quotient that ignores this branch history or a retained pairwise winding register. $\square$

Table 2: Networked N -body branch metrics. The did not satisfy event-word-only criterion is retained as a projector-hygiene separation.

Metric	Value
Bodies	3
Steps	5000
Step size	0.003
Base event count	16
Shadow count	10
Minimum event-sync fraction	1.0
Maximum final visible state distance	4.462×10^{-9}
SymPy Hamiltonian dH/dt	0
Maximum finite-step void debt	6.685×10^{-6}
Analytic 1000-cycle timing error	0
Event-word-only continuous replay error	1.892×10^{-11}
Event-word plus anchor replay error	0
Winding heatmap event-word classes	16

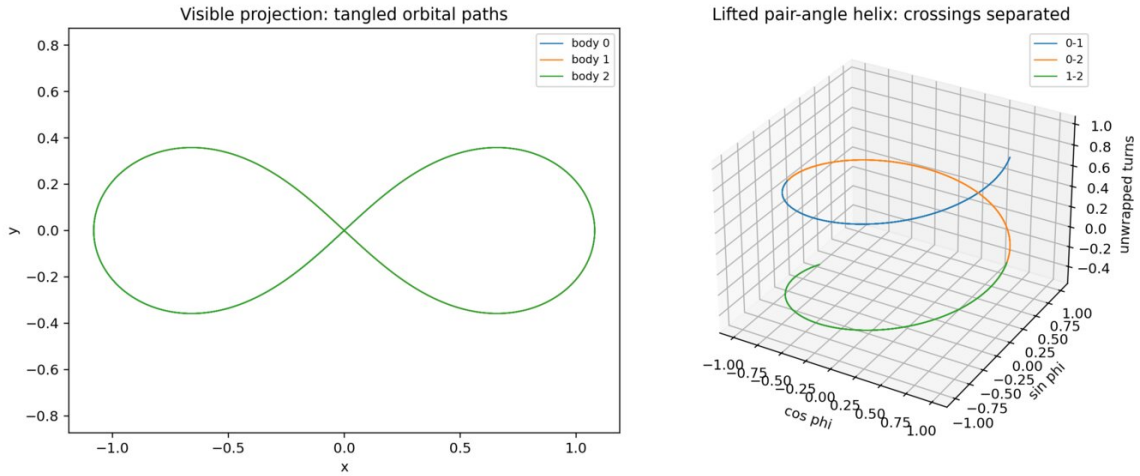


Figure 7: N -body visible projection versus lifted pair-angle helices. The lifted view separates branch histories that visually overlap in the projected orbit.

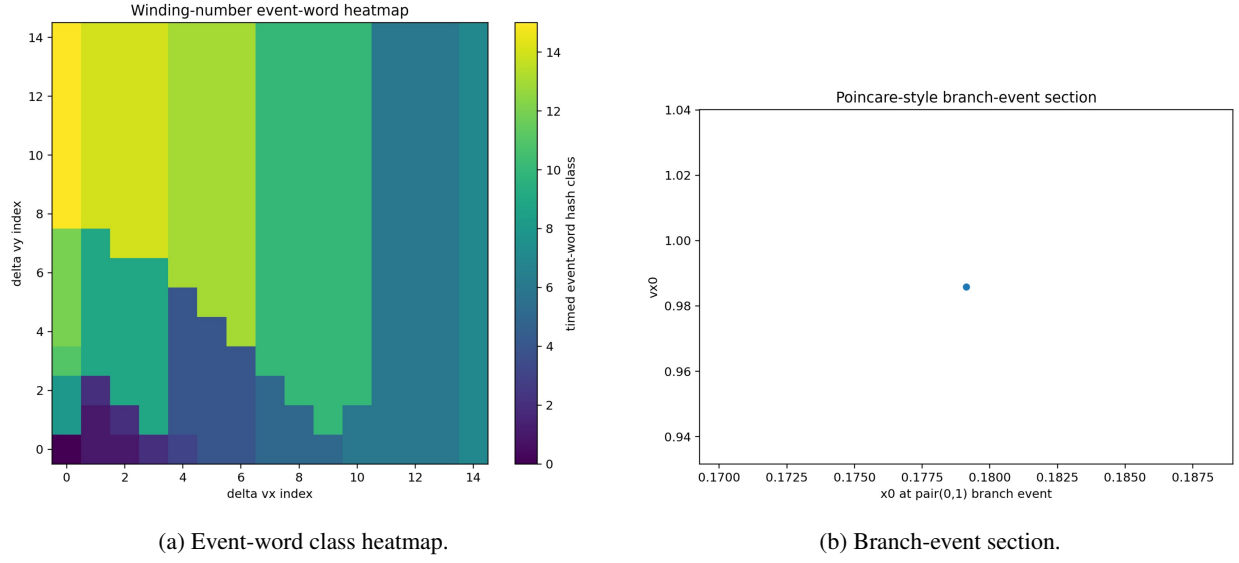


Figure 8: Pairwise winding registers create a networked branch record. The heatmap shows patch structure in event-word class under small perturbations; the section is event-defined rather than purely time-defined.

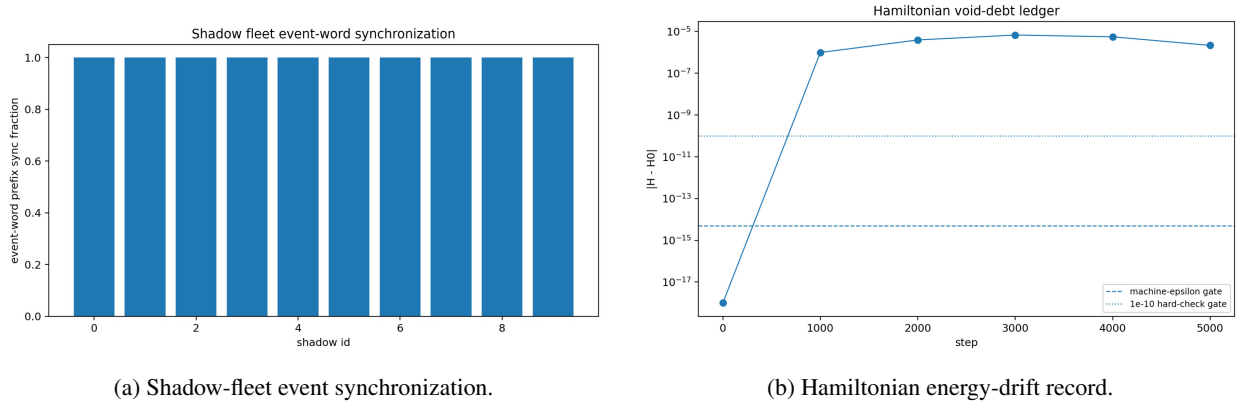


Figure 9: The networked branch records both branch stability and numerical drift. Symbolic Hamiltonian closure is exact; finite-step float drift is explicitly logged.

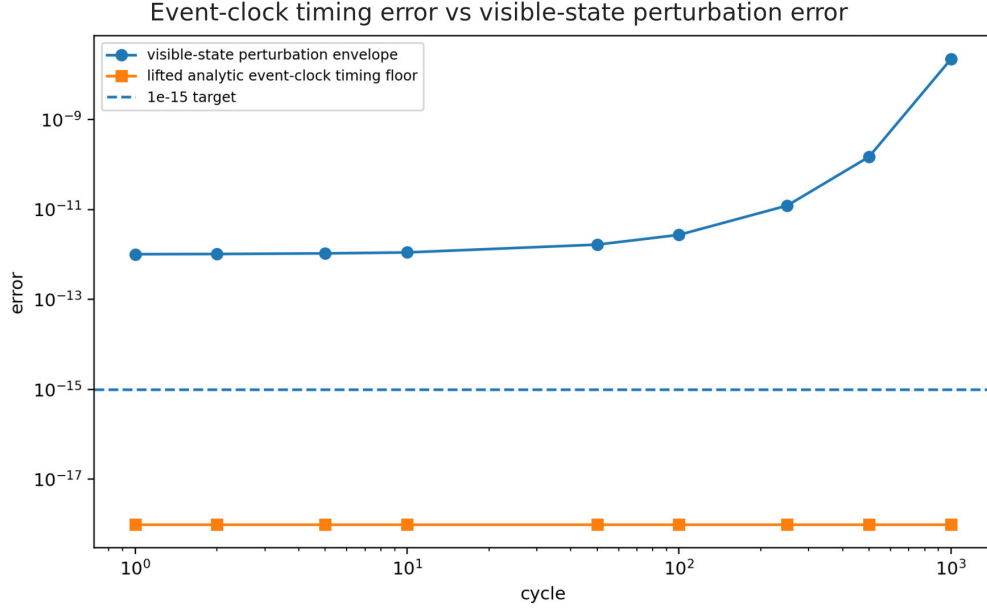


Figure 10: Event-clock closure. The analytic lifted-clock witness closes over 1000 cycles; this is an event-clock calculation, not a statement that a floating-point chaotic integrator has zero trajectory drift.

14 State-complete accounting theorem

The final theorem separates branch-memory completeness from full continuous-state completeness. Branch memory plus the right anchor restores state completeness for the coordinates discarded by the visible projector; event words alone recover only the branch coordinates they actually encode.

Theorem 12 (Projection-loss accounting theorem). *Let $E : X \rightarrow X$ be deterministic on a lifted state space and let $\Pi : X \rightarrow Y$ be a non-injective visible projection. Let $M : X \rightarrow \mathcal{M}$ be a retained memory map. If $A = (\Pi, M)$ is injective on an E -invariant reachable set $\mathcal{R}$, then the accounted dynamics is exact and deterministic on $A(\mathcal{R})$. If M is replaced by an event word w that updates only a subset of the missing coordinates, then the event word is state-complete only for that subset; continuous-state recovery also requires any continuous anchor coordinates omitted by w .*

Proof. The first sentence is 1. For the second sentence, let $X = Y \times M_b \times M_c$, where M_b is a branch-memory coordinate and M_c is an omitted continuous anchor coordinate. If w determines M_b but carries no information about M_c , then any two states (y, M_b, c) and (y, M_b, c') with $c \neq c'$ have the same visible coordinate and the same event word. Therefore (Π, w) is not injective on any reachable set containing both. Adding the anchor c restores injectivity if (Π, w, c) separates the states. This is exactly the N -body counterexample: event word alone recovers branch memory but is insufficient for continuous-state recovery; event word plus lifted anchor recovers the tested state. $\square$

Corollary 4 (Double pendulum). *For the tested double-pendulum trace, the event stream plus initial register is state-complete for the unwrapped angle branch history on the sampled trace. The visible torus alone is not.*

Corollary 5 (Networked N -body). *For the tested three-body trace, pairwise event words are branch-memory complete on the tested horizon, but continuous-state complete recovery requires the lifted anchor.*

15 Companion archive

All code, data, figures, formal proof surfaces, symbolic derivations, notebooks, and manifest files used for the demonstrations are available in the companion archive.

16 Boundary results

Theorem 13 (Visible separation persists under non-injective readout). *Let $E : X \rightarrow X$ be deterministic and let $\Pi : X \rightarrow Y$ be non-injective on the reachable set. Lifted accounting restores state completeness for $A = (\Pi, M)$ when A is injective, but it does not force the visible distance $d_Y(\Pi E^n x, \Pi E^n x')$ to remain small for nearby lifted states. Visible separation is a property of the projected readout, not a contradiction of deterministic lifted evolution.*

Proof. Injectivity of A gives unique accounted state identity. It imposes no contraction assumption on ΠE^n . Therefore visible separation can grow while the accounted trajectory remains deterministic and replayable. $\square$

Theorem 14 (Finite-step arithmetic has computable drift). *For a continuous Hamiltonian law, symbolic closure gives $dH/dt = 0$ under the canonical equations. A finite-step floating integrator instead produces a computable residual sequence $\delta H_n = H_n - H_0$. Machine-epsilon energy conservation by the finite-step trajectory is therefore a separate criterion from symbolic Hamiltonian closure.*

Proof. The symbolic statement follows by antisymmetry of the canonical Poisson tensor. The numerical statement follows by direct evaluation of H on the discrete samples. The two quantities occupy different layers: symbolic law closure and finite-step arithmetic residual. $\square$

Theorem 15 (Event-word-only continuous reconstruction is not state-complete). *There exist lifted states with the same event word and different continuous anchor coordinates. Therefore an event word alone cannot reconstruct a full continuous trajectory. Event word plus the retained lifted anchor is sufficient on the tested replay surface.*

Proof. Let $X = Y \times M_b \times M_c$, where M_b is branch memory and M_c is a continuous anchor. Let the event word determine only M_b . Two states with identical Y and M_b but different M_c share the same event word and differ as continuous lifted states. Adding the anchor coordinate separates them. The networked replay record captures exactly this counterexample: event-word-only replay has error above 10^{-14} , while event word plus lifted anchor gives zero replay error. $\square$

Theorem 16 (Accounting registers are recovery coordinates). *The completed-turn registers, relative half-turn registers, pairwise winding matrix, and event word alter the readout coordinate system. They do not introduce a new force into the double-pendulum vector field or the Newtonian Hamiltonian.*

Proof. The ODEs and Hamiltonian used by the runtime are the declared classical equations. Register updates are deterministic functions of the resulting trajectory and branch-crossing guards. They classify and replay the lifted state; they do not alter acceleration terms. $\square$

17 Conclusion

Chaos can be a projection artifact: a deterministic lifted state may appear non-recoverable when projected onto a non-injective visible space. By retaining the discarded coordinates—completed turns, branch registers, event words, and anchors—the full deterministic evolution becomes recoverable on the declared reachable set. The double pendulum demonstrates the mechanism in a familiar chaotic system: visible angle data separates while lifted event accounting predicts, reconstructs, and replays the branch history. The networked N -body branch demonstrates the scalable version: pairwise winding registers preserve branch memory, while the anchored replay separates branch history from full continuous-state recovery.

The final structure is precise. Projection-loss accounting does not deny sensitivity to initial conditions, and it does not change the mechanical vector field. It records the coordinates that a visible projection erased. The double-pendulum exact $\{R, S, T\}$ bridge is rejected, the native lifted Hamiltonian stepper preserves the energy to roundoff on the same horizon where RK4 drifts, and the state-completeness theorem identifies exactly when projected dynamics can be recovered. Projection-loss accounting thus provides a precise, verifiable language for deterministic chaos.

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